KEMAL KILICASLAN

SUPER INTELLIGENCE DEVELOPER

CONTACT

kemalkilicaslan@outlook.com kemalkilicaslan.com linkedin.com/in/kemalkilicaslan github.com/kemalkilicaslan +905432018373

SUMMARY

Builds projects end to end, from data collection and model training to deployment. Mathematics graduate of Kastamonu University (2022) with 20 production-level projects in ADAS, traffic analytics, medical imaging, fraud detection and on-premise RAG.

SKILLS

Languages: Python, C++, R, SQL (PostgreSQL), MATLAB, Wolfram Mathematica, JavaScript, HTML5, CSS, Swift, LaTeX
Frameworks and tools: PyTorch, TensorFlow, scikit-learn, OpenCV, FFmpeg, PySpark, Docker

EDUCATION

KASTAMONU UNIVERSITY FACULTY OF ARTS AND SCIENCES Mathematics, Bachelor's Degree, 2022 Graduate

EXPERIENCE

AI Content Creator & Brand Partnership **ULTRALYTICS** Mar 2026 - Jul 2026
Produced a sponsored LinkedIn post and a YouTube tutorial for the Ultralytics Platform launch that show the full computer vision pipeline (annotation, training, export, deployment) with YOLO26 on two original real-world datasets, delivered in compliance with FTC guidelines and Ultralytics Brand Guidelines.

Computer Vision Developer **CALORIN** Aug 2024 - Sep 2024
Developed a real-time food volume estimation system with YOLOv8 and OpenCV that segments fruits and vegetables at pixel level and turns the segmented areas into calorie estimates through geometric calculations, with a low-latency inference pipeline validated across different lighting and scale conditions.

Artificial Intelligence Instructor **Bilgi Teknolojileri ve İletişim Kurumu** Jul 2024
I gave more than 20 participants face-to-face artificial intelligence training for a week, starting with Python and concluding with machine learning, deep learning and computer vision. Within the scope of the applied curriculum, TensorFlow, OpenCV and GPU accelerated model training demos were used and participants were directed to develop all stages from data preparation to implementation in an end-to-end manner. The training was completed at a 100% rate with each participant attending all five days.

PROJECTS

Safety Lane Violation Detection System
Real-time detection and monitoring of the emergency lane with YOLOv12. A vehicle that crosses the safety boundary raises an alert and updates the violation count, and color-coded overlays mark each violator for inspection. kemalkilicaslan.com/Safety-Lane-Violation-Detection-System.html

Vehicle Distance Measurement System
Real-time distance measurement with YOLOv12 for in-vehicle cameras. A perspective geometry method measures how far nearby vehicles are in three lanes (left, front and right), adjusts the warning thresholds as conditions change, blurs license plates and shows color-coded safety warnings. kemalkilicaslan.com/Vehicle-Distance-Measurement-System.html

Road Surface Obstacle and Traffic Warning Object Segmentation System
Real-time, pixel-by-pixel segmentation of road hazards, obstacles and traffic warning signs from an in-vehicle camera with the YOLO26x-seg model, so that autonomous vehicles and ADAS systems see more clearly what is in front of them. kemalkilicaslan.com/Road-Surface-Obstacle-and-Traffic-Warning-Object-Segmentation-System.html

Industrial Safety Gear Detection System
Real-time PPE checks with a custom YOLO26x model trained on the Ultralytics Platform. Color-coded overlays show for each worker whether the helmet, safety vest, gloves and face mask are on, at 97.1% mAP50 with 94.6% precision and recall. kemalkilicaslan.com/Industrial-Safety-Gear-Detection-System.html

Vehicle and Pedestrian Traffic Analysis System
Zone-aware traffic analysis with a YOLO26x model trained on the VisDrone dataset and exported to ONNX, and a BoT-SORT tracker written from scratch in C++. A live dashboard analyzes people and vehicles in the crosswalk, road and sidewalk zones. kemalkilicaslan.com/Vehicle-and-Pedestrian-Traffic-Analysis-System.html

Vehicle Plates Detection and Reading System
Real-time license plate detection with a custom-trained, transformer-based RF-DETR model and plate reading with Fast Plate OCR. Each reading is checked against the Turkish plate format, and the results are saved to a labeled video for AVPR. kemalkilicaslan.com/Vehicle-Plates-Detection-and-Reading-System.html

Real-Time Colonoscopy Polyp Segmentation System
Real-time polyp segmentation for colonoscopy video with a PyTorch U-Net, a frame-quality check and an IoU tracker that raises an alarm after three consecutive frames. Training on polyp-free frames cut false alarms on normal colon from 30.6 to 6.2 per minute, and the model scores 0.853 Dice on data from another hospital. kemalkilicaslan.com/Real-Time-Colonoscopy-Polyp-Segmentation-System.html

On-Premise RAG Document Assistant System
On-premise RAG assistant in which an open-weight LLM (Qwen2.5 on Ollama) answers Turkish questions on internal PDFs and cites pages. Semantic search runs on bge-m3 embeddings in pgvector, role-based access control limits users to allowed documents, and 83.7% of TQuAD answers are correct when the right chunk is retrieved. kemalkilicaslan.com/On-Premise-RAG-Document-Assistant-System.html

Parking Lot Occupancy and Vehicle Trajectory Prediction System
Parking lot occupancy analysis and vehicle trajectory prediction for drone footage. It finds 400 slots from the painted lines and marks them occupied or empty with a VisDrone-trained YOLO model. Cars in the aisles are tracked with an OC-SORT style tracker, and a dotted line shows where each will be in the next 2 seconds. kemalkilicaslan.com/Parking-Lot-Occupancy-and-Vehicle-Trajectory-Prediction-System.html

ACTIVITIES

Niğde Ömer Halisdemir Üniversitesi (Speaker) TeknoKonferans NÖHÜ'25 May 2025
Türkiye Matematik Kulübü (Member) Organization team of 6 events that addressed more than 1500 people Nov 2020 - Present
Kastamonu Üniversitesi Matematik & Bilim Topluluğu (Founding President) Speech on 14 March π Day Dec 2021 - Jun 2022

CERTIFICATES

Deep Learning Specialization **DeepLearningAI**
Machine Learning Specialization **Stanford University & DeepLearningAI**
Convolutional Neural Networks in TensorFlow **DeepLearningAI**
Mathematics for Machine Learning and Data Science Specialization **DeepLearningAI**
Self-Driving Cars Specialization **University of Toronto**
Computer Vision Basics **University at Buffalo & The State University of New York**
Machine Learning **Stanford University**